

SoilBR-Onto: An Ontology for Soil Fertility Management and Classification in Brazil

Applied Ontology
2025, Vol. 20(2) 181–196
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DOI: 10.1177/15705838251315571
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Abstract

Soil analysis laboratories require extensive data for precision agriculture, including parameters such as texture, organic matter, phosphorus, and potassium, which guide fertilization and soil acidity correction through pH. This article presents SoilBR-Onto, an ontology that advances the use of fertilizers, liming recommendations and classification of Brazilian soils based on laboratory analysis. A total of 481 axioms and 145 semantic rules enable accurate queries and inferences using real soil sample results. The SoilViewer Mobile application, developed within this study, facilitates access to fertilization and liming recommendations and enables the visualization of soil deficiencies through heat maps. Furthermore, this ontology improves soil data management and decision-making, applicable beyond applied tobacco, to crops such as soybeans, wheat and corn, promoting environmental sustainability and efficiency in agriculture.

Keywords

ontology, context awareness, soil fertility, productivity, precision agriculture

Received: 29 November 2023; accepted: 3 January 2025

1 Introduction

The notion of precision farming involves managing a crop area by monitoring the productivity of specific points, as determined through soil analysis. This data is then used to apply fertilizer and make corrections where necessary. This approach not only results in increased productivity and more efficient use of fertilizer, but it also minimizes the waste of resources. By reducing input waste, this technique helps to prevent soil acidification and salinization, which can eventually compromise the quality of groundwater. Additionally, it can prevent the eutrophication of lakes and rivers caused by leaching (Hochman et al., 2013).

The existence of organic substance or organic carbon within the soil is significant, not only for farming purposes but also for global climate regulation. It aids in the gathering of biomass, filtration and retention of water, recycling of nutrients and provision of a habitat for soil biological activities (Wiesmeier et al., 2019). The quantity of organic matter acts as an indicator of nitrogen availability within the soil, which is crucial in determining the amount of fertilization required for different types of crops (Brazilian Soil Chemistry and Fertility Commission, 2016).

The proportion of clay in the soil establishes the soil's texture, which affects several factors such as plant growth and output, soil water retention, infiltration and transport, nutrient accessibility and uptake by plants, soil quality, temperature and productivity, as well as the levels of soil structure and compaction, irrigation, soil preparation and fertilizer efficiency. As such, the amount of clay present in the soil is crucial in determining crop yield (Tümsavay et al., 2019).

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Table 1. Concentration Ranges of Organic Matter and Clay.

Organic matter		Clay	
Range (%)	Level	Range (%)	Class
≤ 2.5	Low	≤ 20	1
2.6 – 5.0	Medium	21 – 40	2
> 5.0	High	41 – 60	3
		> 60	4

Organic matter and clay in the soil are fundamental in precision agriculture, as these components directly influence crop health and productivity. The combination of these elements allows for more precise adjustments in agricultural practices, such as fertilization, irrigation, and soil preparation, according to the specific needs of each crop and their variations within a field. Both are analyzed in soil testing laboratories to provide detailed information about their presence and levels (Sarkar et al., 2018).

Soil analysis labs require substantial amounts of data to produce reports that are collated and analyzed to assist farmers in determining how to proceed with regards to the use of agricultural inputs (Muñoz & Kravchenko, 2011). Technological advancements in agriculture have facilitated the decision-making process when it comes to managing and utilizing soil, as well as generating vast amounts of data on both soil and plants (de Souza et al., 2010). This data is obtained directly from the fields and labs, necessitating appropriate interpretation strategies to enhance productivity and environmental conservation.

Ontologies, in the context of the Semantic Web, are collections of information that formally define the relationships among terms and concepts within a specific domain. They consist of a taxonomy that categorizes objects and their relationships and a set of inference rules to deduce new information. Ontologies address the issue of different databases using different identifiers for the same concept by providing a shared understanding of terms, enabling more effective data integration and manipulation. They enhance web functionality by improving search accuracy, supporting complex queries, and facilitating data interoperability. Although ontologies do not store instance-level data, they provide a structured and semantically rich framework for organizing and interpreting information, making them a powerful tool for applications requiring precise and meaningful data handling (Berners-Lee et al., 2001).

During the development of an ontology, a common understanding of a specific domain is sought, in practice typically consisting of a collection of entities, connections, functions, axioms and examples. Using operators such as “and,” “or,” and “not” on attributes and relationships, you can restrict cardinalities and configure data ranges. In this way, any individual that meets the logical expression is deduced as a member of this concept. The OWL language can represent ontologies and has the ability to support semantic interoperability, allowing the exchange and sharing of knowledge and information between different systems (Avelino & Silva, 2020; Du et al., 2016; Gualberto et al., 2013; Muñoz et al., 2017).

In this context, ontologies can offer an user-friendly and effective approach to managing soil data, providing assistance for automation and computational intelligence. The AgroRDF proposal, developed by the AgroXML consortium, promotes a data exchange format widely embraced in the agricultural sector, as well as consolidating concepts in line with global requirements (Doluschitz et al., 2005). However, there is a shortage of this type of system for classifying Brazilian soils, primarily due to the existence of over 300 soil analysis laboratories in Brazil (Brazilian Society of Soil Science, 2019, 2020; Embrapa Solos, 2016), as well as a growing demand for precision agriculture analyses.

The Brazilian Society of Soil Science (SBCS) defines the contents of organic matter (O.M.) in three levels (low, medium and high) and clay in four classes, as shown in Table 1 (Helfer et al., 2021).

According to Silva and Tadra (2017), the mixture of sand, silt, and clay in various proportions defines the soil’s textural categories. In Brazil, SBCS adopts 13 different categories of soil texture, including heavy clay, clay, sandy clay, silt clay, sandy clay loam, clay loam, silt clay loam, sandy loam, loam, silt loam, sand, loamy sand, and silt. These types are all visualized in Table 2, which represents a Textural Triangle with ternary concentrations (dos Santos et al., 2005).

The Brazilian regulatory agency responsible for overseeing agricultural activities has a normative instruction that outlines the classification of Brazilian soil types. It utilizes the following criteria for soil classification:

- Type 1: Sandy textured soils with a clay content of at least 10% and less than 15%, or with a clay content equal to or exceeding 15%, in which the difference between the percentage of sand and clay is greater than or equal to 50%.
- Type 2: Medium-textured soils with a clay content of at least 15% and less than 35%, in which the difference between the percentage of sand and clay is less than 50
- Type 3: Clay textured soils with a clay content exceeding 35%.

Table 2. Soil Textural Classification in Brazil.

Type class	Ranges (%) of sand (d), clay (y) and silt (t)
Heavy clay	$y \geq 60$
Clay	$(\leq 40 \ y \leq 60) \wedge d \leq 45 \wedge t < 40$
Sandy clay	$y \geq 35 \wedge d > 45$
Silty clay	$y \geq 40 \wedge t \geq 40$
Sandy clay loam	$(20 \leq y < 35) \wedge t < 28 \wedge d > 45$
Clay loam	$(27 \leq y < 40) \wedge (20 < d \leq 45)$
Silty clay loam	$(27 \leq y < 40) \wedge (d \leq 20)$
Sandy loam	$((7 \leq y < 20) \wedge d > 52 \wedge ((t + 2 * y) \geq 30)) \vee ((y < 7 \wedge t < 50 \wedge (t + 2 * y) \geq 30))$
Loam	$(7 \leq y < 27) \wedge (28 \leq t < 50) \wedge d \leq 52$
Silt loam	$(t \geq 50 \wedge (12 \leq y < 27)) \vee ((50 \leq t < 80) \wedge y < 12)$
Sand	$(t + 1.5 * y) < 15$
Loamy sand	$(t + 1.5 * y \geq 15) \wedge (t + 2 * y < 30)$
Silt	$t \geq 80 \wedge y < 12$

Hence, the aim of this study is to put forward an ontology named SoilBR-Onto that expands on the current understanding of soil classification based on the composition of organic matter and texture. The significance of this endeavor is evident in the lack of a structured ontology for the soil domain, which can function as an essential component of various information systems, such as agricultural management systems, environmental monitoring platforms, and geospatial information systems (Arnaud et al., 2020). A first, much simpler version of this ontology, named Tellus-Onto, was presented at the Brazilian Symposium on Information Systems in 2021, addressing only soil classification (Helfer et al., 2021).

This article is organized into six parts. Section 2 outlines the related studies in the field. Section 3 elucidates the proposed model's methodology. Section 4 showcases the ontology's implementation, while section 5 demonstrates the results of the validations. The final section is also dedicated to concluding remarks.

2 Related Work

The chosen papers focus on the topics of ontology, web semantics, agriculture, and soil. The following articles have been selected and are briefly summarized below.

Muñoz et al. (2017) proposed an ontology for selecting and categorizing 458 soils in Australia based on physical and chemical attributes such as particle size, density, electrical conductivity, and organic carbon. By implementing a *reasoning* process, certain soil orders (chromosol, sodosol, and vertosol) were efficiently and accurately classified, with accuracy exceeding 80% in most cases. The findings demonstrate the potential of ontology-based approaches to efficiently and effectively select samples from vast libraries.

Andrés et al. (2017) developed an ontology-based prototype to automatically categorize terrains using satellite imagery (Landsat) by applying explicit spectral rules. The ontology was structured modularly to create a versatile and generic representation of concepts related to remote sensing. The findings confirmed the potential of ontologies to formalize expert knowledge and classify remote sensing images. However, processing time emerged as a significant limitation.

Helfer et al. (2020) utilized the AgroRDF framework (iGreen Projekt, 2012) to develop modules, agents, and database tables without utilizing the ontology for queries or inferences. Instead, it was used for representing entities and their relationships in a broad sense with the aim of implementing a computational model.

Deb et al. (2020) created a web-based software with N-tier architecture, utilizing an ontology for queries. The ontology was expanded to include the family and series levels, as well as all twelve orders of the United States Department of Agriculture's (USDA) soil taxonomy. The software's research module enables access to soil taxonomy data, while the editing module provides users with the capability to add, delete, and edit ontology information.

Giakoumatos and Gkionakis (2021) developed a knowledge network based on an ontology of the physical-chemical and biological properties of soil/water, derived from ASTM standard methods and scientific references. The research aimed to bridge the gap between the cited standards and the conceptual vocabulary of the infiltration discipline. The soil/water ontology's concepts correspond to their physical-chemical and biological properties, all located in the process of contaminated water infiltration through porous soil media. According to the authors, this approach offers significant advantages to semantic web developers and users, such as rapid knowledge browsing, tracking, retrieval, and usage.

Elumalai and Anuncia (2021) developed a knowledge model for storing and managing information about diverse soil compositions. Their model utilizes an ontology-based approach to represent mineral composition and includes a recommendation system for agricultural soil. The study demonstrated the effectiveness of the model by using 26 semantic

Table 3. Comparison of Related Works: (a) Muñoz et al. (2017), (b) Andrés et al. (2017), (c) Helfer et al. (2020), (d) Deb et al. (2020), (e) Giakoumatos and Gkionakis (2021), (f) Elumalai and Anuncia (2021), (g) Shankhdhar et al. (2021) and (h) this work.

Articles	Inferences	SPARQL	Equivalence	Source	Validation
(a)	Yes	Yes	AGROVOC	Soils from Australia	Real samples
(b)	Yes	No	No	Satellite soil images	Real samples
(c)	No	No	AgroRDF	Soil, and climate analysis	No
(d)	No	No	USDA Soil Taxonomy Ontology	Soil, taxonomy	Real samples
(e)	No	No	No	Soil and water	No
(f)	Yes	No	No	Soils from India	Real samples
(g)	Yes	No	No	IoT data	No
(h)	Yes	Yes	AgroRDF	Soils from Brazil	Real samples

USDA: United States Department of Agriculture.

rules to achieve an accuracy rate of 91%. The authors suggested that accuracy could be enhanced by adding more rules related to knowledge premises and inference, and by incorporating additional relevant facts in the research domain.

Shankhdhar et al. (2021) created an ontological model, SAGRO-Lite, as a part of a multi-agent system called ABSMSA for use in Smart Farming. ABSMSA utilizes the IoT-Lite metamodel for improving performance and features an intelligent agricultural ontology tailored for use by farmers in developing countries, such as India. Semantic modeling is only one aspect of the project. SAGRO-Lite is responsible for managing how the models are used and how annotated data is indexed and queried in real-time with its agents. A specialized ontology named Complex Event Service Ontology manages all events triggered by sensors.

Table 3 illustrates the comparison between the proposed ontology model and related works. To compare the studies and identify significant features for evaluation, the following criteria were established:

- Inferences: It checks whether the inference process is used in the proposed ontology;
- SPARQL: It checks whether the ontology query language is used;
- Equivalence: Checks if the ontology uses an already published database such as AgroRDF, AGROVOC or others;
- Source: It checks the source of soil representation classes;
- Validation: It checks whether tested with real or simulated data.

No literature was found that directly addressed the intersection of ontology and soil classification in precision agriculture. Previous studies that employed ontology to establish relationships and concepts within the soil domain aimed to achieve interoperability between diverse agricultural systems. Other works utilized ontologies solely to facilitate the visualization of entities and their relationships for the implementation of a computational architecture.

The studies conducted by Muñoz et al. (2017) and Elumalai and Anuncia (2021) exhibit the most similarities with the proposed ontology model. While the development of the ontology model is well-documented, the domain and purpose of the methodology developed differs significantly, although it can be regarded as complementary. Muñoz et al. explored soil orders and classifications based on sediment accumulation in Australian soil (Ltd, 2021). Elumalai and Anuncia (2021) classify soils based on their mineralogy and several soil analysis parameters.

3 Methods

An ontology is a formal representation of concepts within a specific knowledge domain, enabling the discovery of answers to questions related to that domain (Grimm et al., 2011). SoilBR-Onto was created using the Protégé software (v. 5.5) in the RDF language (Musen, 2015a). Its concepts and relationships were devised to support and expand upon the AgroRDF pattern maintained by the AgroXML consortium (Doluschitz et al., 2005). The adoption of AgroRDF aligns with the ontology's principle of extensibility, allowing for extensions and specializations without the need for a theory revision.

3.1 Domain Representation Through SoilBR-Onto

During the ontology development process in the Protégé software, each soil type was modeled as disjoint within each group of texture and clay classes, organic matter levels, phosphorus and potassium contents. Therefore, each soil sample could be categorized only once within each group, such as, for example, silty clay, clay class 1, medium organic matter content, low phosphorus content and very low potassium content.

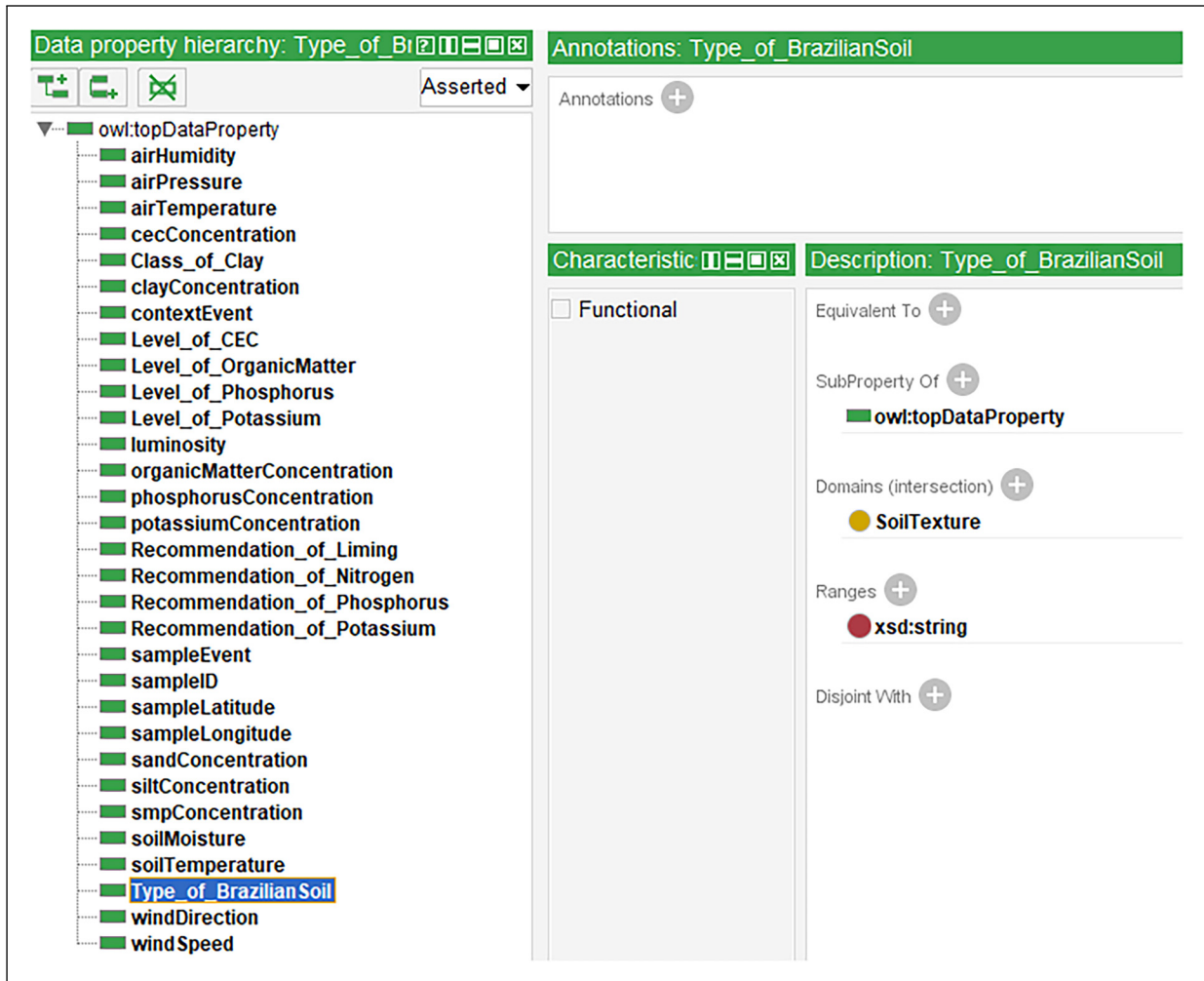


Figure 2. Data properties of SoilBR-Onto.

Axioms are logical expressions that establish a concept. Object properties demonstrate relationships between two class instances, while data properties establish relationships between class instances and literal data types. The “Class count” and “SubclassOf” metrics indicate the number of elements present in the ontology, while the “Individual count” indicates the instances populated for inferences and SPARQL queries. These instances consist of soil samples imported from laboratory reports.

Figure 2 illustrates all data properties developed for SoilBR-Onto.

3.2 Semantic Rules

The soil classification, fertilization, and liming recommendations described in this section were developed using Semantic Web Rule Language (SWRL) definitions, as the calculations involved are complex. SWRL enables users to create rules that can infer new knowledge about individuals (instances) in order to reason about them.

Rules in SWRL consist of two main components: The antecedent (also known as the body) and the consequent (also known as the head). Each rule is an implication that specifies that when the antecedent conditions are satisfied, the consequent conditions are also satisfied.

Table 1 provides the classification of soil organic matter into 3 levels and clay into 4 classes based on their concentrations. This information can be expressed in SWRL rules, such as the example shown in Table 5, which states that “Any concentration of organic matter greater than 2.5% and less than or equal to 5% is classified as medium level.” Similar rules are defined for the low and high levels based on their respective concentration intervals.

Table 5. Semantic Web Rule Language (SWRL) Rule to Represent Organic Matter Medium Level and Clay Class 3.

Rule name	SWRL expression
OM_Medium_Rule	SoilOrganicMatter(?o) $\wedge$ organicMatterConcentration(?o, ?om) $\wedge$ swrlb:greaterThan(?om, 2.5) $\wedge$ swrlb:lessThanOrEqual(?om, 5.0) $\rightarrow$ Level_of_OrganicMatter(?o, "Medium")
Clay_Rule3	ClayTexture(?ct) $\wedge$ clayConcentration(?ct, ?clay) $\wedge$ swrlb:greaterThan(?clay, 20) $\wedge$ swrlb:lessThanOrEqual(?clay, 40) $\rightarrow$ Class_of_Clay(?ct, "Class3")

Table 6. Semantic Web Rule Language (SWRL) Rule to Represent the Category of Sand Type.

Rule name	SWRL expression
Sand_Rule	SoilTexture(?t) $\wedge$ clayConcentration(?t, ?clay) $\wedge$ siltConcentration(?t, ?silt) $\wedge$ swrlb:multiply(?m, ?clay, 1.5) $\wedge$ swrlb:add(?result, ?silt, ?m) $\wedge$ swrlb:lessThan(?result, 15) $\rightarrow$ Category_of_BrazilianSoil(?t, "Sand")

Table 7. Semantic Web Rule Language (SWRL) Rule to Recommend the Amount of Phosphorus Needed (kg of P_2O_5 /ha).

Rule name	SWRL expression
Low_P_Recomm_Rule	ClayTexture(?c) $\wedge$ clayConcentration(?c, ?clay) $\wedge$ swrlb:greaterThan(?clay, 60) $\wedge$ SoilPhosphorus(?p) $\wedge$ phosphorusConcentration(?p, ?pc) $\wedge$ swrlb:lessThanOrEqual(?pc, 1.5) $\rightarrow$ Recommendation_of_Phosphorus(?p, "120")

In the case of clay, a similar approach was used to define rules for its classification. For example, the semantic rule for "Class 3" type is shown in Table 5 where "Any concentration of clay greater than 20% and less than or equal to 40% is classified as Class 3." Similar rules were defined for the other classes based on their respective concentration intervals.

Table 6 displays the rule for the *Sand* category, which is one of the 13 texture categories of Brazilian soil. In this case, a data property named *Category_of_BrazilianSoil* was defined to ensure the inference is performed accurately.

A similar situation occurs in the recommendations. For example, to recommend the required amount of phosphorus, it is first necessary to infer which clay class the sample comprises and inform the amount of phosphorus contained. One of the applied rules can be viewed in Table 7. The expression can be read as "Any instance of Texture *c* that has more than 60 (%) of clay and instance of Phosphorus *p* has its content greater than 1.5 (mg/dm³), infers the recommendation of 120 (kg of P_2O_5 /ha). All semantic rules developed in this ontology are available at <https://github.com/ghelfer/SoilBR/blob/main/swrl.pdf> while the entire file can be found at <https://ghelfer.net/ontologies/soil-ontology/agro.rdf>, both under the W3C open-source license.

4 SoilBR-Onto Implementation

Figure 3 illustrates the architecture of the SoilBR-Onto ontology system. It is composed of an API that acts as a communication channel for querying and configuring the system. Sensor data from telemetry stations is recorded in the database and made available to the ontology. Similarly, georeferenced soil analysis collections and soil laboratory analysis results are also available through the API. Data availability is ensured through SPARQL queries.

The "ontology base" mentioned in the mechanism contains all the classes, subclasses, object and data properties, as well as the semantic rules described in the previous sections of this chapter. Two inference configuration options are available: one static and one dynamic.

In the static configuration, the user needs to manually request the reasoner to generate a new ontology with inferences from the created *Individuals* in the database. This new ontology is then published for queries and represents the open and connected data model. Although it offers faster query response times, the manual reasoner process may result in non-updated data.

In the dynamic configuration, every query request triggers a reasoner process, which generates a new ontology based on the database *Individuals*. This new ontology is published for queries and represents an updated and connected model.

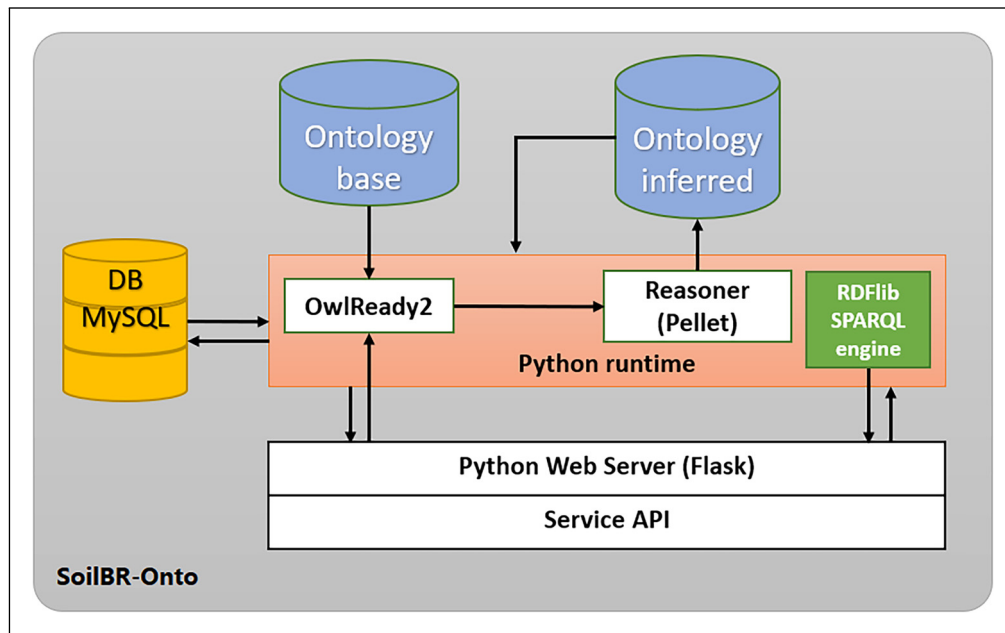


Figure 3. SoilBR-Onto's mechanism.

Table 8. Types of Soil in Brazil.

Soil type	Rules for clay and sand concentrations
Type 1	$10\% \leq \text{clay} < 15\%$
Type 1	$\text{clay} < 10\% \wedge \text{sand} \geq 60\%$
Type 1	$\text{clay} \geq 15\% \wedge (\text{sand} - \text{clay}) \geq 50\%$
Type 1	$\text{sand} \geq 90\%$
Type 2	$15\% \leq \text{clay} < 35\% \wedge (\text{sand} - \text{clay}) < 50\%$
Type 3	$\text{clay} \geq 35\%$

The advantage of this configuration is the constant updating of the data, although it may involve longer processing times for each query request.

Regarding the ontology, an application called SoilViewer Mobile has the function of informing the soil type, soil classifications and recommending the necessary amount of fertilizers and acidity correction as a result of the inferences made. Figure 4 illustrates the interfaces of the application, such as login (a) and a heat map representing the potassium recommendation (b) of the working area.

4.1 Soil Classification by Inference

Besides representing knowledge, SoilBR-Onto performs the inference of soil type and recommendations for fertilization (application of NPK fertilizer) and liming (application of limestone). It also classifies the soil in terms of organic matter levels, clay classes and the 13 categories of the Textural Triangle.

The soil type classification follows the regulations of the Brazilian Agriculture Agency and is also contained in SoilBR-Onto, and its rules follow the specification shown in Table 8.

4.2 Fertilization Recommendation by Inference

The recommendations used as a standard for this study follow the Liming and Fertilization Manual for Southern Region in Brazil, published by Brazilian Soil Science Society. This manual is periodically revised by various sectors of the production chain, universities and the agronomic community (SBCS, 2016).



Figure 4. Application login screen (a) and sampling region (b).

Table 9. Nitrogen (N) Recommendation per Hectare.

Tobacco type O.M.%	Virginia kg of N/ha	Burley kg of N/ha
≤1,0	160 – 180	190 – 210
1,1 – 2,0	140 – 160	180 – 190
2,1 – 3,0	120 – 140	160 – 180
3,1 – 4,0	110 – 120	140 – 160
4,1 – 5,0	100 – 110	120 – 140
<5,0	≤100	≤120

The determination of nitrogen recommendations is based on the measurement of soil organic matter conducted by the laboratory analysis, and specific recommendations are determined for each crop. For Virginia tobacco (*Nicotiana tabacum*), recommended N values are based on the guidelines outlined in Table 9.

The recommended interpretation of phosphorus content relies on the classification of the clay and the P parameter (mg/dm³) provided by soil analysis reports. It is also specific to each crop. In the case of Virginia tobacco, for instance, the recommendations for phosphorous pentoxide values are presented in Table 10.

The recommendations for potassium oxide contents depend on the CEC values (Cation Exchange Capacity) at pH7 in cmol_c/dm³, as well as the K parameter (mg/dm³) found in the soil analysis reports, and the specific crop to be cultivated. Virginia tobacco, for instance, has its potassium oxide recommendation values based on the guidelines provided in Table 11.

Table 10. Phosphorus (P_2O_5) Recommendation per Hectare.

Clay	Class 1	Class 2	Class 3	Class 4	
P level	Concentration of Phosphorus (mg/dm^3)				kg of P_2O_5/ha
Very low	≤ 1.5	≤ 2.0	≤ 3.0	≤ 5.0	120
Low	1.6 – 3.0	2.1 – 4.0	3.1 – 6.0	5.1 – 10.0	80
Medium	3.1 – 4.5	4.1 – 6.0	6.1 – 9.0	10.1 – 15.0	60
High	4.6 – 9.0	6.1 – 12.0	9.1 – 18.0	15.1 – 30.0	40
Very high	> 9.0	> 12.0	> 18.0	> 30.0	≤ 20

Table 11. Potassium (K_2O) Recommendation per Hectare.

CEC _{pH7}	≤ 7.5	7.6 – 15	15.1 – 30.0	> 30.0	
K level	Concentration of Potassium (mg/dm^3)				kg of K_2O/ha
Very low	≤ 15	≤ 20	≤ 30	≤ 35	200
Low	16 – 30	21 – 40	31 – 60	36 – 70	160
Medium	31 – 45	41 – 60	61 – 90	71 – 105	140
High	46 – 90	61 – 120	91 – 180	106 – 210	120
Very high	> 90	> 120	> 180	> 210	≤ 105

Table 12. Liming Recommendation (tons of $CaCO_3/ha$).

SMP	Desirable pH			SMP	Desirable pH		
	5.5	6.0	6.5		5.5	6.0	6.5
≤ 4.4	15.0	21.0	29.0	5.8	2.3	4.2	6.3
4.5	12.5	17.3	24.0	5.9	2.0	3.7	5.6
4.6	10.9	15.1	20.0	6.0	1.6	3.2	4.9
4.7	9.6	13.3	17.5	6.1	1.3	2.7	4.3
4.8	8.5	11.9	15.7	6.2	1.0	2.2	3.7
4.9	7.7	10.7	14.2	6.3	0.8	1.8	3.1
5.0	6.6	9.9	13.3	6.4	0.6	1.4	2.6
5.1	6.0	9.1	12.3	6.5	0.4	1.1	2.1
5.2	5.3	8.3	11.3	6.6	0.2	0.8	1.6
5.3	4.8	7.5	10.4	6.7	0	0.5	1.2
5.4	4.2	6.8	9.5	6.8	0	0.3	0.8
5.5	3.7	6.1	8.6	6.9	0	0.2	0.5
5.6	3.2	5.4	7.8	7.0	0	0	0.2
5.7	2.8	4.8	7.0	7.1	0	0	0

4.3 Liming Recommendation by Inference

The liming recommendation aims to increase the soil pH to the desired level in a single application, which varies depending on the type of crop to be grown and the target pH range, usually between 5.5 and 6.5. For Virginia tobacco, the recommended pH is 6.0 based on the SMP index of the soil analyzed by the laboratory. Table 12 presents the complete recommended amount of $CaCO_3$ tons per hectare.

4.4 Findable, Accessible, Interoperable, Reusable (FAIR) Principles Compliance

The SoilBR-Onto ontology developed in this study aligns with the 20 FAIR principles, ensuring greater accessibility, interoperability, and reusability of data in the field of agriculture and soil science (Wilkinson et al., 2016). This approach strengthens the manuscript's position by demonstrating that the work not only addresses technical challenges in knowledge representation but also establishes a quality standard that can serve as a reference for future developments in the area.

4.4.1 Findable. The ontology is designed to be easily identifiable and accessible through structured metadata, ensuring that the concepts and axioms utilized are well-documented and searchable. The integration of tools like the SoilViewer

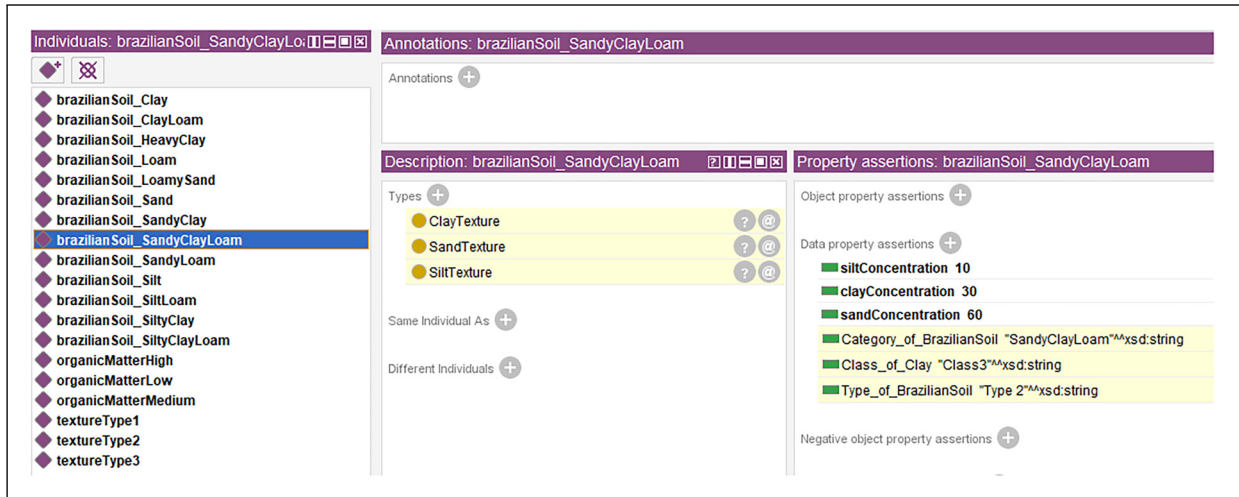


Figure 5. List of manually generated instances for SoilBR-Onto testing.

Mobile app reinforces this principle by providing a practical interface for retrieving information related to fertilization and liming recommendations.

4.4.2 Accessible. Accessibility is enhanced by making the ontology publicly available in a standardized format, enabling laboratories, researchers, and farmers to access data through platforms like the mobile app. Furthermore, the semantic infrastructure employs open standards, such as OWL and SPARQL, ensuring access across diverse systems, independent of proprietary platforms.

4.4.3 Interoperable. The project promotes interoperability by leveraging widely accepted semantic standards, such as RDF and SWRL, which allow integration with other ontologies and agricultural databases. This facilitates data sharing across different systems and domains, ensuring that information on soils, fertilizers, and agricultural practices can be consistently understood and applied in various contexts.

4.4.4 Reusable. Reusability is central to the design of SoilBR-Onto. Its hierarchical and connected structure is intended for use in various crops beyond those tested in the study, such as soybeans, wheat, and corn. The axioms and semantic rules, thoroughly documented and validated with real data, provide a replicable model that can be adapted and expanded by other researchers and institutions.

4.4.5 Impact on the Scientific Community and the Field. By adhering to the FAIR principles, SoilBR-Onto contributes to improving the quality and reliability of agricultural data representation while offering a reusable foundation for new studies. The availability of a robust ontological model not only strengthens data management in the agricultural sector but also inspires the use of ontologies as powerful tools to address global challenges related to sustainability and efficiency.

Applying the FAIR principles adds significant value to the development of the ontology and positions this work as a milestone in the field of soil science and agricultural management, encouraging the adoption of similar practices in future projects.

5 Results

Initially, a set of instances was created to cover the full range of soil types and classifications, which was used for evaluation purposes. Figure 5 displays the manually generated individuals, and an inference example.

Afterwards, 15 results of soil samples from an analysis laboratory collected in a tobacco field were added as instances to the ontology. The semantic rules described in the previous section were processed by the Pellet reasoner, which is a well-known ontology reasoner for the RDF/OWL language. Pellet was accessed through the OwlReady2 Python module.

Figure 6 illustrates the inference process performed on a soil sample identified as *s03-T* and manually added according to report number 195551/21 (Figure 7). The inference process classified this instance as having *Low* organic matter content, *Class 4* clay, *Type 1* soil, and *SandyLoam* texture.

s03 - T — <https://ghelfer.net/ontologies/soil-ontology#195551/21>

Annotations: s03 - T

Annotations:
 rdfs:label [type: xsd:string]
 s03 - T

Description: s03 - T

Property assertions: s03 - T

Types:
 ClayTexture
 Liming
 Nitrogen
 Phosphorus
 Potassium
 SandTexture
 SiltTexture
 Soil
 SoilCEC
 SoilOrganicMatter
 SoilPhosphorus
 SoilPotassium

Object property assertions:
 clayConcentration 18
 siltConcentration 11
 smpConcentration 7.1
 organicMatterConcentration 2.0
 phosphorusConcentration 94.1
 sandConcentration 71
 cecConcentration 7.9
 potassiumConcentration 478
 Category_of_BrazilianSoil "SandyLoam"
 Class_of_Clay "Class4"
 Level_of_CEC "Medium"
 Level_of_OrganicMatter "Low"
 Level_of_Phosphorus "Very High"
 Level_of_Potassium "Very High"
 Recommendation_of_Liming "0"
 Recommendation_of_Nitrogen "140-160"
 Recommendation_of_Phosphorus "<=20"
 Recommendation_of_Potassium "<= 105"
 Type_of_BrazilianSoil "Type 1"

Figure 6. Soil sample ID 195551/21 (s03-T) inference result.

Sample	Clay %	Class	pH H ₂ O	SMP index	P mg.dm ⁻³	K mg.dm ⁻³	O.M. %	Al cmol _c .dm ⁻³	Ca cmol _c .dm ⁻³	Mg cmol _c .dm ⁻³
195553/21	18	Class 4	6,6	7,1	94,1	478	2,0	0,0	4,0	1,4

Sample	H + Al cmol _c .dm ⁻³	CEC (cmol _c .dm ⁻³)		% CEC Saturation			Relation Ca/Mg	Relation Ca/K	Relação Mg/K	Relation (Ca+Mg)/K	Sum of Bases (cmol _c .dm ⁻³)
		Effective	pH 7,0	Al	Bases	K					
195553/21	1,2	6,7	7,9	0,0	84,5	15,5	2,8	3,3	1,2	4,4	6,7

Sample	Sample Identification				Hectares		Land Identification	
195553/21	03 - T				-		-	

Sample	PHYSICAL ANALYSIS			
	Clay %	Sand %	Silt %	Classification
195559/21	18	71	11	Type 1

Sample	Sample Identification				Hectares		Land Identification	
195559/21	03 - T				-		-	

Figure 7. Soil analysis report ID 195551/21 (s03-T).

The classification of the soil sample into its respective categories is done based on the analysis report provided by the laboratory and the rules defined in the ontology. The engineer or agronomist responsible for the analysis and interpretation of the report can use the ontology and reasoning process to obtain more accurate and consistent results.

The explanation of this inference is illustrated in Figure 8 which shows its belonging to the domain *SoilOrganicMatter* in addition to the application of the rule *OM_Low_Rule*.

To validate the classification of soil through the Textural Triangle, instances were generated in Protégé using the Cellfie plugin. These instances comprised the results of 98 soil samples obtained from a laboratory. The Pellet reasoner was used to carry out the inference process and all samples were correctly classified. A Python routine based on Table 2 was used to compare the results.

To validate the recommendations for fertilization and liming, an Excel spreadsheet was created to calculate the recommendations based on the soil analysis results. The outputs were exactly the same as the inferences from the ontology. Figure 9 shows the report of sample S03T (b), which had its recommendations calculated by the spreadsheet (a) and the ontology (c).

6 Conclusion

SoilBR-Onto is an ontology designed to provide soil classification, fertilization, and liming recommendations for the Brazilian agricultural sector. The ontology was built to be extensible to AgroRDF and allows for inferences and the representation of entities in the model.

One of the key features of this ontology is its generality, as each country has its own rules regarding textural classes, allowing for new relationships to be established between terms as needed.

Using a reasoner, axioms, and SWRL-based rules, the ontology was able to perform precise and automatic inferences and queries using data from laboratory reports and historical contexts. This study highlights the potential of ontologies in agriculture as an effective tool for decision support platforms.

Using the SoilViewer Mobile application allowed access to fertilization and liming recommendations through the inferences made. In addition to the NPK and Liming results to be used, a heat map helped soil experts to more quickly visualize the places where the soil had a certain deficiency and needed some correction.

The ontology developed in this research represents Brazilian soil knowledge and allows for inference of recommendations for precision agriculture management. Although it was implemented specifically for tobacco, adjustment of parameters in the semantic rules could enable its application to any type of crop in Brazil, such as wheat, rice, and others.

In future work, expanding the ontology to include other crops is a key suggestion, as Brazil is a major exporter of grains such as soybeans, wheat, and corn. This expansion would require only adjustments to the SWRL criteria, given their flexibility to accommodate different crops. An open and connected data scenario could then encompass the entire production chain of these agricultural sectors, streamlining processes and standardizing routines.

However, as SoilBR-Onto grows to cover a broader range of crops and incorporates more detailed regional information, managing and querying the increasing amount of instance-level data may become more complex and resource-intensive, underscoring a need for scalability solutions.

To address these challenges, future efforts will focus on integrating more comprehensive datasets from various regions of Brazil to enhance the ontology's accuracy and relevance. Leveraging machine learning models, such as convolutional neural networks, could enable real-time data processing and inference generation, improving the system's ability to provide precise recommendations.

Additionally, further development of the SoilViewer Mobile application will be prioritized to ensure intuitive access for soil experts and farmers, complemented by training and documentation for end-users. Establishing partnerships with agricultural research institutions and government bodies will also promote the system's credibility and facilitate its adoption on a larger scale.

Alongside these advancements, future revisions will explore the integration of SoilBR-Onto with top-level ontologies, such as the Basic Formal Ontology (BFO), and its alignment with agricultural ontologies like the Agronomy Ontology and Plant Ontology, to enhance semantic rigor and interoperability. Quantitative and qualitative evaluations of SoilBR-Onto's reasoning speed, query efficiency, and integration capabilities compared to related works like SAGRO-Lite and AgroRDF will further highlight its unique contributions and innovative potential in advancing sustainable agricultural practices.

Acknowledgment

The authors acknowledge the Rio Grande do Sul State Research Support Foundation (FAPERGS), the Higher Education Personnel Improvement Coordination - Brazil (CAPES) - Financing Code 001, the National Council for Scientific and Technological Development (CNPq), the University of Santa Cruz do Sul and the Central Analytical Sector, for providing soil samples, as well as the University of

the Vale do Rio dos Sinos (Unisinos) for supporting the development of this work. The authors would like to especially acknowledge the Unisinos Graduate Program in Applied Computing (PPGCA) and the Mobile Computing Laboratory (Mobilab) support.

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Statements and Declarations

Funding

The authors received no financial support for the research, authorship and/or publication of this article.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

References

- Andrés, S., Arvor, D., Mougnot, I., Libourel, T., & Durieux, L. (2017). Ontology-based classification of remote sensing images using spectral rules. *Computers & Geosciences*, 102, 158–166. <https://doi.org/10.1016/j.cageo.2017.02.018>
- Arnaud, E., Laporte, M.-A., Kim, S., Aubert, C., Leonelli, S., Cooper, L., Jaiswal, P., Kruseman, G., Shrestha, R., Buttigieg, P. L., Mungall, C. J., Pietragalla, J., Agbona, A., Muliro, J., Detras, J., Hualla, V., Rathore, A., Das, R., Dieng, I., Bauchet, G., Menda, N., Pommier, C., Shaw, F., Lyon, D., Mwanzia, L., Juarez, H., Bonaiuti, E., chiputwa, b., Obileye, O., Sandrine Auzoux, S. A., Dzalé Yeumo, E., Mueller, L., Silverstein, K., Lafargue, A., Antezana, E., Devare, M., & King, B. (2020). The ontologies community of practice: An initiative by the CGIAR platform for big data in agriculture. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.3565982>
- Avelino, J. G., & Silva, P. D. A. (2020). Ontologia de redes de valor verdes. *iSys - Brazilian Journal of Information Systems*, 13(2), 168–197. <https://doi.org/10.5753/isys.2020.767>
- Berners-Lee, T., Hendler, J., & Lassila, O. (2001). Scientific american: Feature article: The semantic web. A new form of Web content that is meaningful to computers will unleash a revolution of new possibilities. <https://tinyurl.com/S3M4N7IC>.
- Brazilian Society of Soil Science. (2019). CELA - SBCS. <https://sbcs-nepar.org.br/cela>
- Brazilian Society of Soil Science. (2020). SBCS - Núcleo Região Sul. <http://www.sbcs-nrs.org.br/index.php?secao=rolas>
- Brazilian Soil Chemistry and Fertility Commission. (2016). *Manual de adubação e de calagem para os Estados do Rio Grande do Sul e de Santa Catarina* (p.400). Porto Alegre: Brazilian Society of Soil Science.
- Deb, C. K., Marwaha, S., & Pandey, R. N. (2020). Ontology learning algorithm for development of ontologies from taxonomic text and USDA soil taxonomy ontology. *Journal of the Indian Society of Agricultural Statistics*, 1(74), 77–84.
- de Souza, Z. M., Cerri, D. G. P., Colet, M. J., Rodrigues, L. H. A., Magalhães, P. S. G., & Mandoni, R. J. A. (2010). Análise Dos Atributos do Solo e da Produtividade da Cultura de Cana-de-açúcar Com o Uso da Geoestatística E árvore de Decisão. *Ciência Rural*, 40(4), 840–847. <https://doi.org/10.1590/s0103-84782010005000048>
- Doluschitz, R., Kunisch, M., Jungbluth, T., & Eider, C. (2005). agroXML - A standardized data format for information flow in agriculture. In *Proceedings of the EFITA/WCCA 2005 joint conference*. EFITA/WCCA 2005 (pp. 439–443). Universidade de Trás-os-Montes e Alto Douro.
- dos Santos, R., Lemos, R., Santos, H., Ker, J., & dos Anjos, L. (2005). *Manual De Descrição E Coleta De Solo No Campo* (5th ed.). Viçosa-MG: Sociedade Brasileira de Ciência do Solo - Centro Nacional de Pesquisa de Solos.
- Du, H., Dimitrova, V., Magee, D., Stirling, R., Curioni, G., Reeves, H., Clarke, B., & Cohn, A. (2016). An ontology of soil properties and processes. In *Lecture notes in computer science* (pp. 30–37). Springer International Publishing. https://doi.org/10.1007/978-3-319-46547-0_4.
- Elumalai, M., & Anuncia, S. M. (2021). Development of soil mineral classification using ontology mining. *Arabian Journal of Geosciences*, 14(14), <https://doi.org/10.1007/s12517-021-07651-w>
- Embrapa Solos. (2016). Tecnologia inovadora analisa solos em apenas 30 segundos - Portal Embrapa. <https://www.embrapa.br/busca-de-noticias/-/noticia/17162564/tecnologia-inovadora-analisa-solos-em-apenas-30-segundos/>
- Giakoumatos, S. D. V., & Gkionakis, A. K. T. (2021). Development of an ontology-based knowledge network by inter-connecting soil/water concepts/properties, derived from standards methods and published scientific references outlining infiltration/percolation process of contaminated water. *Journal of Geoscience and Environment Protection*, 09(01), 25–52. <https://doi.org/10.4236/gep.2021.91003>
- Grimm, S., Abecker, A., V'olker, J., & Studer, R. (2011). Ontologies and the semantic web. In *Handbook of semantic web technologies* (pp. 507–579). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-540-92913-0_13.
- Gualberto, E. S., Souza Jr., R. T., Deus, F. E. G., & Duque, C. G. (2013). Proposição de uma Ontologia de Apoio à Gestão de Riscos de Segurança da Informação. *iSys – Revista Brasileira de Sistemas de Informação*, 6(1), 30–43.

- Helfer, G. A., Barbosa, J. L. V., dos Santos, R., & da Costa, A. B. (2020). A computational model for soil fertility prediction in ubiquitous agriculture. *Computers and Electronics in Agriculture*, 175, 105602. <https://doi.org/10.1016/j.compag.2020.105602>
- Helfer, G. A., da Costa, A. B., Bavaresco, R. S., & Barbosa, J. L. V. (2021). Tellus-Onto: uma ontologia para classificação e inferência de solos na agricultura de precisão. In *XVII Brazilian symposium on information systems*. ACM. <https://doi.org/10.1145/3466933.3466946>
- Hochman, Z., Carberry, P. S., Robertson, M. J., Gaydon, D. S., Bell, L. W., & McIntosh, P. C. (2013). Prospects for ecological intensification of australian agriculture. *European Journal of Agronomy*, 44, 109–123. <https://doi.org/10.1016/j.eja.2011.11.003>
- iGreen Projekt. (2012). Welcome to the AgroRDF RDFLib LOD server. <http://data.igreen-services.com/agrordf>
- Ltd, S. Q. P. (2021). Measuring soil texture in the laboratory. <http://soilquality.org.au/factsheets/soil-texture-measuring-in-the-lab>
- Muñoz, J. D., & Kravchenko, A. (2011). Soil carbon mapping using on-the-go near infrared spectroscopy, topography and aerial photographs. *Geoderma*, 166(1), 102–110.
- Muñoz, A., Soriano-Disla, J. M., & Janik, L. J. (2017). An ontology-based approach for an efficient selection and classification of soils. *Ambient Intelligence and Smart Environments*, 22(Intelligent Environments 2017), 69–78. <https://doi.org/10.3233/978-1-61499-796-2-69>
- Musen, M. A. (2015a). The protégé project. *AI Matters*, 1(4), 4–12.
- Sarkar, B., Singh, M., Mandal, S., Churchman, G. J., & Bolan, N. S. (2018). Clay minerals-organic matter interactions in relation to carbon stabilization in soils. In *The future of soil carbon* (pp. 71-86-). Elsevier. <https://doi.org/10.1016/b978-0-12-811687-6.00003-1>
- SBCS. (2016). *Manual de calagem e adubação para os Estados do Rio Grande do Sul e de Santa Catarina* (11 ed., p.376). Porto Alegre-RS: Sociedade Brasileira de Ciência do Solo - Núcleo Regional Sul - Comissão de Química e Fertilidade do Solo.
- Shankhdhar, G. K., Sharma, R., & Darbari, M. (2021). SAGRO-Lite: A light weight agent based semantic model for the internet of things for smart agriculture in developing countries. In R. Pandey, M. Paprzycki, N. Srivastava, S. Bhalla, & K. Wasielewska-Michniewska (Eds.), *Semantic IoT: Theory and applications: Interoperability, provenance and beyond* (pp. 265–302). CHAM: Springer International Publishing. https://10.1007/978-3-030-64619-6_12
- Silva, N. M. D., & Tadra, R. M. S. (2017). *Geologia e Pedologia*. Curitiba: InterSaberes. (p.320).
- Tümsavas, Z., Tekin, Y., Ulusoy, Y., & Mouazen, A. M. (2019). Prediction and mapping of soil clay and sand contents using visible and near-infrared spectroscopy. *Biosystems Engineering*, 177, 90–100. *Intelligent Systems for Environmental Applications*. <https://doi.org/10.1016/j.biosystemseng.2018.06.008>
- Wiesmeier, M., Urbanski, L., Hobbey, E., Lang, B., von Lützw, M., Marin-Spiotta, E., van Wesemael, B., Rabot, E., Ließ, M., Garcia-Franco, N., Wollschläger, U., Vogel, H.-J., & Kögel-Knabner, I. (2019). Soil organic carbon storage as a key function of soils - A review of drivers and indicators at various scales. *Geoderma*, 333, 149–162. <https://doi.org/10.1016/j.geoderma.2018.07.026>
- Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., da Silva Santos, L. B., Bourne, P. E., Bouwman, J., Brookes, A. J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C. T., Finkers, R., Gonzalez-Beltran, A., Gray, A. J. G., Groth, P., Goble, C., Grethe, J. S., Heringa, J., 't Hoen, P. A. C., Hooft, R., Kuhn, T., Kok, R., Kok, J., Lusher, S. J., Martone, M. E., Mons, A., Packer, A. L., Persson, B., Rocca-Serra, P., Roos, M., van Schaik, R., Sansone, S.-A., Schultes, E., Sengstag, T., Slater, T., Strawn, G., Swertz, M. A., Thompson, M., van der Lei, J., van Mulligen, E., Velterop, J., Waagmeester, A., Wittenburg, P., Wolstencroft, K., Zhao, J., & Mons, B. (2016). The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3(1). <https://doi.org/10.1038/sdata.2016.18>